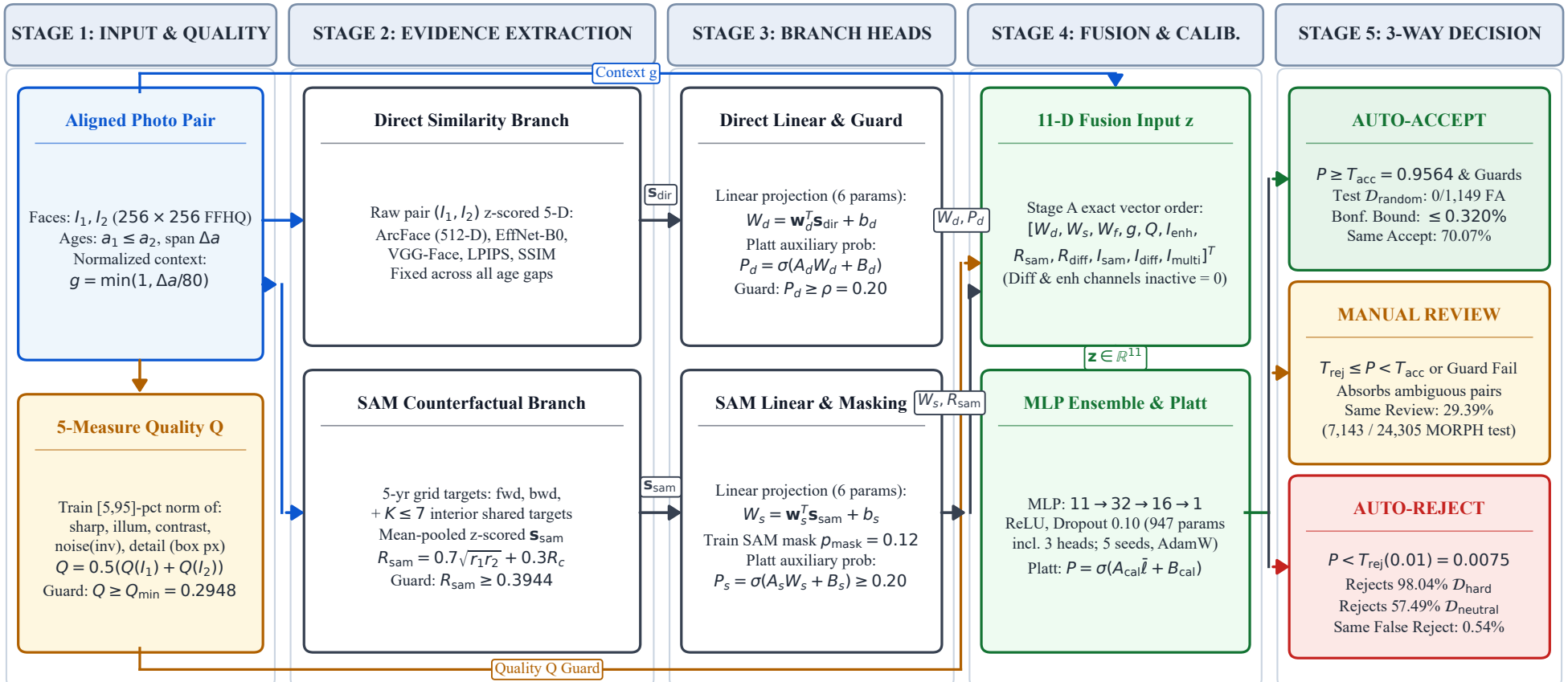

STAGE 1: INPUT & QUALITY
STAGE 2: EVIDENCE EXTRACTION
STAGE 3: BRANCH HEADS
STAGE 4: FUSION & CALIB.
STAGE 5: 3-WAY DECISION
Context g
Aligned Photo Pair
Faces: I_1, I_2 (256 × 256 FFHQ)
Ages: $a_1 \le a_2$, span Δa
Normalized context:
$g = \min(1, \Delta a/80)$
5-Measure Quality Q
Train [5,95]-pct norm of:
sharp, illum, contrast,
noise(inv), detail (box px)
$Q = 0.5(Q(I_1) + Q(I_2))$
Guard: $Q \ge Q_{\min} = 0.2948$
Direct Similarity Branch
Raw pair (I_1, I_2) z-scored 5-D:
ArcFace (512-D), EffNet-B0,
VGG-Face, LPIPS, SSIM
Fixed across all age gaps
SAM Counterfactual Branch
5-yr grid targets: fwd, bwd,
+ $K \le 7$ interior shared targets
Mean-pooled z-scored $\mathbf{s}_{\text{sam}}$
$R_{\text{sam}} = 0.7\sqrt{r_1 r_2} + 0.3R_c$
Guard: $R_{\text{sam}} \ge 0.3944$
$\mathbf{s}_{\text{dir}}$
$\mathbf{s}_{\text{sam}}$
Direct Linear & Guard
Linear projection (6 params):
$W_d = \mathbf{w}_d^T \mathbf{s}_{\text{dir}} + b_d$
Platt auxiliary prob:
$P_d = \sigma(A_d W_d + B_d)$
Guard: $P_d \ge \rho = 0.20$
SAM Linear & Masking
Linear projection (6 params):
$W_s = \mathbf{w}_s^T \mathbf{s}_{\text{sam}} + b_s$
Train SAM mask $p_{\text{mask}} = 0.12$
Platt auxiliary prob:
$P_s = \sigma(A_s W_s + B_s) \ge 0.20$
W_d, P_d
W_s, R_{sam}
11-D Fusion Input z
Stage A exact vector order:
$[W_d, W_s, W_f, g, Q, I_{\text{enh}},$
$R_{\text{sam}}, R_{\text{diff}}, I_{\text{sam}}, I_{\text{diff}}, I_{\text{multi}}]^T$
(Diff & enh channels inactive = 0)
$\mathbf{z} \in \mathbb{R}^{11}$
MLP Ensemble & Platt
MLP: $11 \to 32 \to 16 \to 1$
ReLU, Dropout 0.10 (947 params
incl. 3 heads; 5 seeds, AdamW)
Platt: $P = \sigma(A_{\text{cal}} \bar{\ell} + B_{\text{cal}})$
Quality Q Guard
AUTO-ACCEPT
$P \ge T_{\text{acc}} = 0.9564$ & Guards
Test $\mathcal{D}_{\text{random}}$: 0/1,149 FA
Bonf. Bound: $\le 0.320\%$
Same Accept: 70.07%
MANUAL REVIEW
$T_{\text{rej}} \le P < T_{\text{acc}}$ or Guard Fail
Absorbs ambiguous pairs
Same Review: 29.39%
(7,143 / 24,305 MORPH test)
AUTO-REJECT
$P < T_{\text{rej}}(0.01) = 0.0075$
Rejects 98.04% $\mathcal{D}_{\text{hard}}$
Rejects 57.49% $\mathcal{D}_{\text{neutral}}$
Same False Reject: 0.54%